① Să se determine mulțimea de convergență și suma seriilor de puteri

a) $\sum_{n\ge2}\frac{1}{n}x^{2n}$ $x\in\mathbb{R}$

$R=\frac{1}{\lim_{n\to\infty}\sqrt[n]{\frac{1}{n}}}=\frac{1}{1}=1\Rightarrow I_c\in(-1;1)$

$x=\pm1\Rightarrow\sum_{n\ge2}\frac{1}{n}(\pm1)^{2n}=\sum_{n\ge2}\frac{1}{n}$ diverg $\Rightarrow M_c=(-1;1)$

$S(x)=\sum_{n\ge2}\frac{1}{n}\cdot x^{2n}$ $\Bigg|\Rightarrow f(y)=\sum_{n\ge2}\frac{1}{n}y^n\xRightarrow{\text{deriv}}f'(y)=\sum_{n\ge2}\frac{1}{n}\cdot n\cdot y^{n-1}$

Notăm $y=x^2$ $\Bigg|$ $f'(y)=\sum_{n\ge2}y^{n-1}=\sum_{n\ge1}y^n$

$\sum_{n\ge0}y^n=\frac{1}{1-y}$ pt. $|y|<1\Rightarrow1+\sum_{n\ge1}y^n=\frac{1}{1-y}\Rightarrow\sum_{n\ge1}y^n=\frac{1}{1-y}-1\Bigg|\Rightarrow$

$\Rightarrow f'(y)=\frac{1}{1-y}-1\xRightarrow{\int_0^y}f(y)=\int_0^y\left(\frac{1}{1-t}-1\right)dt=\left[-\ln(1-t)-t\right]\Big|_0^y$

$f(y)=-\ln(1-y)-y$

$S(x)=f(x^2)=-\ln(1-x^2)-x^2$

b) $\sum_{n\ge2}\frac{(-1)^n}{n}\cdot x^{2n}$ $x\in\mathbb{R}$

$R=\frac{1}{\lim_{n\to\infty}\sqrt[n]{\left|\frac{(-1)^n}{n}\right|}}=1\Rightarrow I_c\in(-1;1)$

$x=\pm1\Rightarrow\sum_{n\ge2}\frac{(-1)^n}{n}\cdot(\pm1)^{2n}=\sum_{n\ge2}\frac{(-1)^n}{n}$ $\Bigg|\xRightarrow[\text{Leib.}]{\text{Crit.}}\sum_{n\ge2}\frac{(-1)^n}{n}$ converg.

$\alpha_n=\frac{1}{n}$ $\quad\alpha_n>0$, $\alpha_n\searrow$, $\lim_{n\to\infty}\alpha_n=0$ $\Bigg|\Rightarrow M_c=[-1;1]$

$S(x)=\sum_{n\ge2}\frac{(-1)^n}{n}x^{2n}$ $\Bigg|\Rightarrow f(y)=\sum_{n\ge2}\frac{(-1)^n}{n}\cdot y^n\xRightarrow{\text{deriv.}}f'(y)=\sum_{n\ge2}(-1)^n\cdot y^{n-1}$

Notăm $x^2=y$ $\Bigg|$ $f'(y)=\sum_{n\ge1}(-1)^{n+1}\cdot y^n$

$\sum_{n\ge0}(-1)^n\cdot y^n=\frac{1}{1+y}\,\Big|\cdot(-1)\Rightarrow\sum_{n\ge0}(-1)^{n+1}\cdot y^n=\frac{-1}{1+y}\Rightarrow-1+\sum_{n\ge1}(-1)^{n+1}\cdot y^n=\frac{-1}{1+y}\Bigg|\Rightarrow$

$\sum_{n\ge1}(-1)^{n+1}\cdot y^n=1-\frac{1}{1+y}$

$\Rightarrow f'(y)=1-\frac{1}{1+y}\xRightarrow{\int_0^y}f(y)=\int_0^y\left(1-\frac{1}{1+t}\right)dt=\left[t-\ln(1+t)\right]\Big|_0^y$

$\Rightarrow f(y)=y-\ln(1+y)$

$S(x)=f(x^2)=x^2-\ln(1+x^2)$

c) $\sum_{n\ge1}\frac{1}{2n+1}x^{2n+1}$ $x\in\mathbb{R}$

$a_n=\frac{1}{2n+1}$ $\quad R=\frac{1}{\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|}=\frac{1}{\lim_{n\to\infty}\left|\frac{2n+3}{2n+1}\right|}=1\Rightarrow I_c=(-1;1)$

Pt. $x=1\Rightarrow\sum_{n\ge1}\frac{1}{2n+1}\cdot1^{2n+1}=\sum_{n\ge1}\frac{1}{2n+1}$ $\Bigg|\Rightarrow\sum_{n\ge1}\frac{1}{2n+1}$ diverg

$\sum_{n\ge1}\frac{1}{2n+1}\sim\sum_{n\ge1}\frac{1}{n}$

$\sum_{n\ge1}\frac{1}{n}$ diverg $\qquad\Bigg|\Rightarrow M_c=(-1;1)$

Pt. $x=-1\Rightarrow\sum_{n\ge1}\frac{1}{2n+1}(-1)^{2n+1}=-\sum_{n\ge1}\frac{1}{2n+1}$ $\Bigg|\Rightarrow-\sum_{n\ge1}\frac{1}{2n+1}$ diverg.

cf. celor de mai sus

$S(x)=\sum_{n\ge1}\frac{1}{2n+1}x^{2n+1}\xRightarrow{\text{deriv.}}S'(x)=\sum_{n\ge1}x^{2n}=x^2+x^4+\dots$ $\Bigg|\Rightarrow$

$\sum_{n\ge0}x^n=\frac{1}{1-x}\Rightarrow\sum_{n\ge0}x^{2n}=\frac{1}{1-x^2}\Rightarrow1+\sum_{n\ge1}x^{2n}=\frac{1}{1-x^2}$

$S'(x)=\frac{1}{1-x^2}-1\xRightarrow{\int_0^x}S(x)=\int_0^x\frac{1}{1-t^2}-1\,dt=\int_0^x\frac{1}{2}\left(\frac{1}{1-t}+\frac{1}{1+t}\right)-1\,dt$

$S(x)=\left\{\frac{1}{2}\left[-\ln(1-t)+\ln(1+t)\right]-t\right\}\Big|_0^x=\frac{1}{2}\ln\frac{1+x}{1-x}-x$

d) $\sum_{n\ge1}\frac{(-1)^{n-1}}{n}x^{4n}$, $x\in\mathbb{R}$

$R=\frac{1}{\lim_{n\to\infty}\sqrt{\left|\frac{(-1)^{n-1}}{n}\right|}}=\frac{1}{1}=1\Rightarrow I_c=(-1;1)$

Pt. $x=\pm1\Rightarrow\sum_{n\ge1}\frac{(-1)^{n-1}}{n}\cdot(\pm1)^{4n}=\sum_{n\ge1}\frac{(-1)^{n-1}}{n}$ $\Bigg|\xRightarrow[\text{Leibniz}]{\text{Crit.}}\sum_{n\ge1}\frac{(-1)^{n-1}}{n}$ converg

$a_n=\frac{1}{n}\Rightarrow a_n>0$, $a_n\searrow$, $\lim_{n\to\infty}a_n=0$ $\Bigg|\Rightarrow M_c=[-1;1]$

$S(x)=\sum_{n\ge1}\frac{(-1)^{n-1}}{n}x^{4n}$ $\Bigg|\Rightarrow f(y)=\sum_{n\ge1}\frac{(-1)^{n-1}}{n}\cdot y^n\xRightarrow{\text{deriv.}}f'(y)=\sum_{n\ge1}(-1)^{n-1}\cdot y^{n-1}$

Notăm $y=x^4$ $\Bigg|$ $f'(y)=\sum_{n\ge0}(-1)^n\cdot y^n=\frac{1}{1+y}\xRightarrow{\int_0^y}f(y)=\int_0^y\frac{1}{1+t}dt$

$f(y)=\ln(1+y)\Rightarrow S(x)=\ln(1+x^4)$

② Să se aproximeze cu o eroare mai mică decât 10^{-3} integralele

a) $\int_0^{1/2}\frac{1-\cos x}{x^2}dx$

$\frac{1-\cos x}{x^2}=\frac{1-\sum_{n\ge0}(-1)^n\cdot\frac{x^{2n}}{(2n)!}}{x^2}=\frac{\cancel{1}-\left(\cancel{1}-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+\dots\right)}{x^2}=\frac{\frac{x^2}{2!}-\frac{x^4}{4!}+\frac{x^6}{6!}-\dots}{x^2}=$

$=\frac{1}{2!}-\frac{x^2}{4!}+\frac{x^4}{6!}-\dots=\sum_{n\ge1}(-1)^{n+1}\cdot\frac{x^{2n-2}}{(2n)!}$

$\int_0^{1/2}\frac{1-\cos x}{x^2}dx=\int_0^{1/2}\left(\sum_{n\ge1}(-1)^{n+1}\cdot\frac{x^{2n-2}}{(2n)!}\right)dx=\sum_{n\ge1}\frac{(-1)^{n+1}}{(2n)!}\int_0^{1/2}x^{2n-2}dx$

$=\sum_{n\ge1}\frac{(-1)^{n+1}}{(2n)!}\cdot\frac{x^{2n-1}}{2n-1}\Big|_0^{1/2}=\sum_{n\ge1}\frac{(-1)^{n+1}}{(2n)!}\cdot\frac{\left(\frac{1}{2}\right)^{2n-1}}{2n-1}=\sum_{n\ge1}\frac{(-1)^{n+1}}{(2n)!(2n-1)2^{2n-1}}$

$\left|\frac{(-1)^{n+1}}{(2n)!}\cdot\frac{1}{2^{2n-1}(2n-1)}\right|<10^{-3}\Rightarrow(2n)!\,2^{2n-1}(2n-1)>1000$

Prima valoare pt. care inegalitatea este adevărată e $\boxed{n=3}\Rightarrow S\simeq S_2$

$S\simeq S_2=\frac{(-1)^2}{2!\cdot1\cdot2^1}+\frac{(-1)^3}{4!\cdot3\cdot2^3}$

b) $\int_0^1\frac{1-e^{-x^2}}{x^2}dx$

$\frac{1-e^{-x^2}}{x^2}=\frac{\cancel{1}-\left(\cancel{1}-\frac{x^2}{2!}+\frac{x^4}{3!}-\frac{x^6}{4!}+\dots\right)}{x^2}=\frac{1}{2!}-\frac{x^2}{3!}+\frac{x^4}{4!}-\dots=\sum_{n\ge1}(-1)^{n+1}\cdot\frac{x^{2n-2}}{n!}$

$\xRightarrow{\int_0^1}\sum_{n\ge1}\frac{(-1)^{n+1}}{n!}\cdot\int_0^1x^{2n-2}dx=\sum_{n\ge1}\frac{(-1)^{n+1}}{n!}\cdot\frac{x^{2n-1}}{2n-1}\Big|_0^1=\sum_{n\ge1}\frac{(-1)^{n+1}}{n!}\cdot\frac{1}{2n-1}$

$\left|\frac{(-1)^{n+1}}{n!}\cdot\frac{1}{(2n-1)}\right|<10^{-3}\Rightarrow n!(2n-1)>1000$

$n=4\Rightarrow4!\cdot7>1000\Rightarrow168>1000$ fals

$n=5\Rightarrow5!\cdot9>1000\Rightarrow1080>1000$ adev.

$n=5$ Ⓐ $\Rightarrow S\simeq S_4\Rightarrow S=\frac{1}{1!}-\frac{1}{2!\cdot3}+\frac{1}{3!\cdot5}-\frac{1}{4!\cdot7}$

c) $\int_0^{1/2}\frac{\sin x}{x}dx$

$\frac{\sin x}{x}=\frac{\sum_{n\ge0}\frac{(-1)^n}{(2n+1)!}x^{2n+1}}{x}=\sum_{n\ge0}\frac{(-1)^n}{(2n+1)!}x^{2n}$

$\int_0^{1/2}\frac{\sin x}{x}dx=\sum_{n\ge0}\frac{(-1)^n}{(2n+1)!}\int_0^{1/2}x^{2n}dx=\sum_{n\ge0}\frac{(-1)^n}{(2n+1)!}\cdot\frac{x^{2n+1}}{2n+1}\Big|_0^{1/2}=$

$=\sum_{n\ge0}\frac{(-1)^n}{(2n+1)!}\cdot\frac{\left(\frac{1}{2}\right)^{2n+1}}{2n+1}=\sum_{n\ge0}\frac{(-1)^n}{2^{2n+1}\cdot(2n+1)!(2n+1)}$

$\left|\frac{(-1)^n}{2^{2n+1}(2n+1)!(2n+1)}\right|<10^{-3}\Rightarrow2^{2n+1}(2n+1)!(2n+1)>1000$

$n=1\Rightarrow2^3\cdot3!\cdot3>1000$ fals

$n=2\Rightarrow2^5\cdot5!\cdot5>1000$ adev.

$n=2$ Ⓐ $\Rightarrow S\simeq S_1\Rightarrow S=\frac{1}{2\cdot1!\cdot1}-\frac{1}{2^3\cdot3!\cdot3}$

d) $\int_0^{1/2}\frac{\ln(1+x)}{x}dx$

$\frac{\ln(1+x)}{x}=\frac{\sum_{n\ge1}\frac{(-1)^{n+1}}{n}x^n}{x}=\sum_{n\ge1}\frac{(-1)^{n+1}}{n}\cdot x^{n-1}$

$\int_0^{1/2}\frac{\ln(1+x)}{x}dx=\sum_{n\ge1}\frac{(-1)^{n+1}}{n}\int_0^{1/2}x^{n-1}dx=\sum_{n\ge1}\frac{(-1)^{n+1}}{n}\frac{x^n}{n}\Big|_0^{1/2}=\sum_{n\ge1}\frac{(-1)^{n+1}}{n}\cdot\frac{\left(\frac{1}{2}\right)^n}{n}$

$=\sum_{n\ge1}\frac{(-1)^{n+1}}{2^n\cdot n^2}$

$\left|\frac{(-1)^{n+1}}{2^n\cdot n^2}\right|<10^{-3}\Rightarrow2^n\cdot n^2>1000$

$n=5\Rightarrow2^5\cdot5^2>1000\Rightarrow32\cdot25>1000$ fals

$n=6\Rightarrow2^6\cdot6^2>1000\Rightarrow64\cdot36>1000$ adev.

$n=6$ Ⓐ $\Rightarrow S\simeq S_5\Rightarrow S=\frac{1}{2\cdot1^2}-\frac{1}{2^2\cdot2^2}+\frac{1}{2^3\cdot3^2}-\frac{1}{2^4\cdot4^2}+\frac{1}{2^5\cdot5^2}$

e) $\int_0^{1/3}\frac{\operatorname{arctg}x}{x}dx$

$\frac{\operatorname{arctg}x}{x}=\frac{\sum_{n\ge0}\frac{(-1)^n}{2n+1}\cdot x^{2n+1}}{x}=\sum_{n\ge0}\frac{(-1)^n}{2n+1}\cdot x^{2n}$

$\int_0^{1/3}\frac{\operatorname{arctg}x}{x}dx=\sum_{n\ge0}\frac{(-1)^n}{2n+1}\int_0^{1/3}x^{2n}dx=\sum_{n\ge0}\frac{(-1)^n}{2n+1}\,\frac{x^{2n-1}}{2n-1}\Big|_0^{1/3}=$

$=\sum_{n\ge0}\frac{(-1)^n}{2n+1}\cdot\frac{\left(\frac{1}{3}\right)^{2n-1}}{2n-1}=\sum_{n\ge0}\frac{(-1)^n}{(2n+1)(2n-1)3^{2n-1}}$

$\left|\frac{(-1)^n}{(2n+1)(2n-1)3^{2n-1}}\right|<10^{-3}\Rightarrow(2n+1)(2n-1)3^{2n-1}>1000$

$n=2\Rightarrow5\cdot3\cdot3^3>1000\Rightarrow15\cdot27>1000$ fals

$n=3\Rightarrow7\cdot5\cdot3^5>1000\Rightarrow35\cdot243>1000$ adev.

$n=3$ adev $\Rightarrow S\simeq S_2\Rightarrow S\simeq\frac{1}{1\cdot(-1)\cdot3^{-1}}+\frac{-1}{3\cdot1\cdot3}+\frac{1}{5\cdot3\cdot3^3}$

f) $\int_0^1e^{-x^2}dx$

$e^{-x^2}=\sum_{n\ge0}\frac{1}{n!}\cdot(-x^2)^n=\sum_{n\ge0}\frac{(-1)^n}{n!}x^{2n}$

$\int_0^1e^{-x^2}dx=\sum_{n\ge0}\frac{(-1)^n}{n!}\int_0^1x^{2n}=\sum_{n\ge0}\frac{(-1)^n}{n!}\cdot\frac{x^{2n+1}}{2n+1}\Big|_0^1=\sum_{n\ge0}\frac{(-1)^n}{n!(2n+1)}$

$\left|\frac{(-1)^n}{n!(2n+1)}\right|<10^{-3}\Rightarrow n!(2n+1)>1000$

$n=4\Rightarrow4!\cdot9>1000\Rightarrow$ fals

$n=5\Rightarrow5!\cdot11>1000\Rightarrow$ adev.

$n=5$ adev $\Rightarrow S\simeq S_4\Rightarrow S\simeq\frac{1}{0!\cdot1}+\frac{(-1)}{1!\cdot3}+\frac{1}{2!\cdot5}+\frac{(-1)}{3!\cdot7}+\frac{1}{4!\cdot9}$

③ Să se scrie polinoamele Taylor de ordinul 1 și 2 pt. funcțiile următoare în jurul punctelor indicate:

a) $f(x)=1+x\ln x$; $a=1$ să se calculeze aproximativ $\ln\frac{11}{10}$ cu ajutorul pol. T_2

$f'(x)=\ln x+x\cdot\frac{1}{x}=1+\ln x$ $\qquad f(1)=1+0=1$

$f''(x)=\frac{1}{x}$ $\qquad f'(1)=1+0=1$

$f''(1)=\frac{1}{1}=1$

$T_1=f(a)+\frac{f'(a)}{1!}(x-a)=\cancel{1}+(x-\cancel{1})=x$

$T_2=f(a)+\frac{f'(a)}{1!}(x-a)+\frac{f''(a)}{2!}(x-a)^2=\cancel{1}+(x-\cancel{1})+\frac{1}{2}(x-1)^2$

$=x+\frac{1}{2}(x^2-2x+1)=\cancel{x}+\frac{1}{2}x^2-\cancel{x}+\frac{1}{2}=\frac{1}{2}(x^2+1)$

$\ln\frac{11}{10}=\alpha=?$

$f\left(\frac{11}{10}\right)\simeq T_2\left(\frac{11}{10}\right)\Rightarrow1+\frac{11}{10}\ln\frac{11}{10}\simeq\frac{1}{2}\left[\left(\frac{11}{10}\right)^2+1\right]\Rightarrow$

$\Rightarrow\frac{11}{10}\alpha\simeq\frac{1}{2}+\frac{121}{200}-1=\frac{121}{200}-\frac{1}{2}=\frac{121-100}{200}=\frac{21}{200}\Rightarrow\alpha=\frac{21\cdot1\cancel{0}}{11\cdot20\cancel{0}}$

$\alpha=\ln\left(\frac{11}{10}\right)\simeq\frac{21}{11\cdot20}$

b) $f(x)=(x+1)e^x$; $a=0$ să se aproximeze $e^{-0{,}1}$ cu ajutorul polinomului T_2.

$f'(x)=e^x+(x+1)e^x=e^x(x+2)$ $\qquad f(0)=1$

$f''(x)=e^x(x+2)+e^x=e^x(x+3)$ $\qquad f'(0)=2$

$f''(0)=3$

$T_1=f(a)+\frac{f'(a)}{1!}(x-a)=1+\frac{2}{1!}x=1+2x$

$T_2=f(a)+\frac{f'(a)}{1!}(x-a)+\frac{f''(a)}{2!}(x-a)^2=1+2x+\frac{3}{2!}x^2$

$T_2=1+2x+\frac{3}{2}x^2$

$f(-0{,}1)\simeq T_2(-0{,}1)=1+2(-0{,}1)+\frac{3}{2}(-0{,}1)^2=1-0{,}2+\frac{0{,}03}{2}$

$(-0{,}1+1)e^{-0{,}1}=0{,}8+0{,}015\Rightarrow e^{-0{,}1}=-0{,}1+0{,}015$